THE MALAŴI NATIONAL EXAMINATIONS BOARD

2005 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

MATHEMATICS

Subject Number: M131/II

Time Allowed: 2 h 30 mins

Monday, 24 October

8:30 – 11:00 am

PAPER II

(100 marks)

Instructions

1. **This paper contains 6 pages. Please check.**
2. Answer **all** the **six** questions in Section **A** and any **three** questions from Section **B**.
3. The maximum number of marks for each answer is indicated against each question.
4. Mathematical tables, graph paper and answer books are provided.
5. Calculators may be used.
6. Used graph paper and/or supplementary sheets must be tied together inside the answer book with a string.
7. **All working must be clearly shown**; it should be done on the same sheet as the rest of the answers.
8. Write your **Examination Number** on top of each page of your Answer Book.

© 2005 MANEB

Section A (55 marks)

Answer **all** the **six** questions in this section.

1. a. Simplify to the lowest term,

$$\frac{x-2}{x^2-3x+2} \div \frac{x}{x-1}.$$ **(3 marks)**

b. Express $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ with a rational denominator in its simplest form. **(5 marks)**

2. a. Make n the subject of the formula $\log y^n = 2$. **(3 marks)**

b. Calculate the values of a and b if $\begin{pmatrix} 3a & 18 \\ 4 & 6 \end{pmatrix} - \begin{pmatrix} 2a & 2 \\ 2 & -6 \end{pmatrix} = \begin{pmatrix} 4 & 16 \\ 2 & b \end{pmatrix}$. **(4 marks)**

3. a. **Figure 1** shows a square **DEFB** of side 5 cm inside triangle **ABC**.

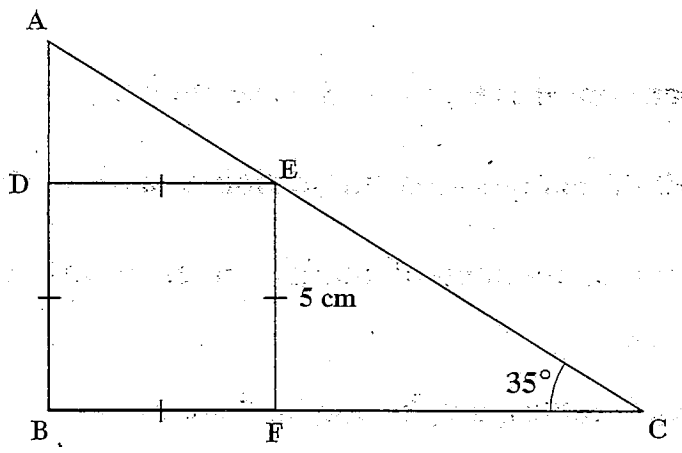

Figure 1

If angle **ACB** = 35°, calculate the length of **AB**. **(5 marks)**

b. Find the remainder when $t^3 + 2t^2 - 3(t + 1)$ is divided by $(t + 1)$. **(4 marks)**

Continued/...

4. a. In **Figure 2**, **XYZ** is a circle with centre **O**. **TXP** is a tangent to the circle at **X**, and the diameter **ZY** produced meets the tangent at **T**.

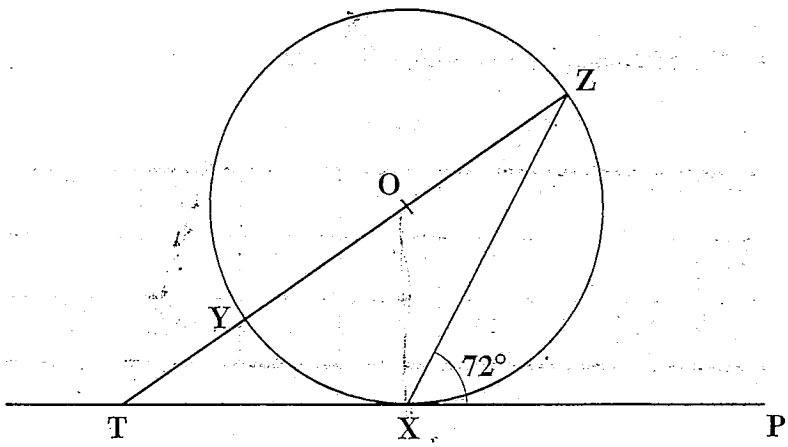

Figure 2

If angle **ZXP** = 72°, calculate the value of the angle **XTY**. **(5 marks).**

b. The function $f(x) = \dfrac{2x-1}{x}$ is defined on the domain $\{-1, \frac{1}{2}, 1\}$.
Draw the arrow diagram to represent this function. **(5 marks)**

5. a. A metallic sphere of volume 770 cm^3 is melted and made into a solid cylinder of length 5 cm. Calculate the radius of the cylinder. (Take $\pi = \frac{22}{7}$) **(4 marks)**

b. A cyclist starts from rest and accelerates uniformly for 4 minutes to reach a speed of 300 metres/minute. She then maintains this speed for 6 minutes after which she decelerates uniformly for 5 minutes to come to a complete stop.

Using a scale of 2 cm to represent 50 metres/minute in the vertical axis and 2 cm to represent 2 minutes in the horizontal axis, draw a speed-time graph for the cyclist on the graph paper provided. **(7 marks)**

6. a. Two triangles **LMN** and **ABC** are similar. In triangle **LMN**, **LM** = 4 cm, **MN** = 5 cm and **LN** = 6 cm. The longest side of triangle **ABC** is 5 cm longer than its shortest side. Find the ratio of the areas of the two triangles. **(6 marks)**

b. Let $\overrightarrow{AB} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$. Calculate the mid-point of $\overrightarrow{BC}$.

(4 marks)

Continued/...

Section B (45 marks)

Answer any **three** questions from this section.

7. a. **Table 1** shows the results of a test which 30 pupils sat for.

Table 1

2	9	14	7	12	3	19	7	13	19
8	14	23	7	18	12	9	14	8	22
17	9	12	18	14	18	13	12	24	4

(i) Using the class intervals of the marks as 1-5, 6-10, 11-15, 16-20, 21-25, construct a frequency table for the marks.

(ii) Using your frequency table, draw a frequency polygon. **(8 marks)**

b. A rectangular garden has a perimeter of 40 metres. If its area is 91 m^2, calculate the length of the garden. **(7 marks)**

8. a. The cost (C) for an international call from Malawi to Europe, partly varies inversely as time, *(t)*, and partly as the square of the time.

A one-minute call costs K120 and a two-minute call costs K200. Find the cost of a five-minute call. **(8 marks)**

b. A family has 3 children born at different times. Assuming it is equally likely to have a baby boy or a baby girl,

(i) draw a tree diagram to show the probabilities of having a boy or a girl on each of the three births.

(ii) calculate the probability that the family has 2 boys and 1 girl in any order of births. **(7 marks)**

9. a. Solve for x if $(2^x)^2 - 9(2^x) + 8 = 0$. **(5 marks)**

b. A city assembly decides to construct a 1400 m^2 car park for lorries and minibuses. A minibus will fit on a 10 m^2 space while a lorry requires 15 m^2 of space.

The number of lorries has to be greater or equal to half the number of minibuses. The number of lorries has to be less than twice the number of minibuses.

(i) If x represents the number of minibuses and y represents the number of lorries, write down **one** inequality involving x and y in addition to $y \geq \frac{x}{2}$ and $y < 2x$.

$10x + 15y \leq 1400$

Continued/...

9. b. (Continued)

(ii) Using a scale of 2 cm to represent 20 units on both axes, draw the region **R** bounded by the **three** inequalities.

(iii) Use your graph to find the maximum number of vehicles (minibuses and lorries) that can be parked on a car park of such a size. **(10 marks)**

10. a. A straight line passes through points (–1, –2) and (3,4). Find the equation of the straight line in the form $y = mx + c$. **(5 marks)**

b. The ratio of the 2nd term to the 7th term of an arithmetic progression is 1:3 and their sum is 20. Calculate the sum of the first 10 terms of the progression. **(10 marks)**

11. a. **Figure 3** shows two unequal circles **PRS** and **QRS** intersecting at **R** and **S**. **TP** and **TQ** are tangents to the circles at **P** and **Q** respectively. **PR** and **RQ** are diameters of the circles.

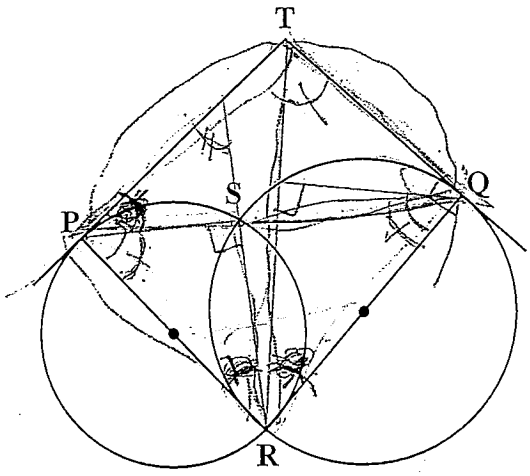

Figure 3

Prove that:

(i) **PRQT** is cyclic quadrilateral;

(ii) angle **PRS** = angle **QRT**. **(8 marks)**

b. Given that **A, B** and **C** are sets,

(i) draw a venn diagram and shade the region representing $A' \cap B' \cap C$.

(ii) find $n(A' \cap B' \cap C)$, if $n(A \cup B) = 8$ and $n(A \cup B \cup C) = 12$.

(7 marks)

Continued/...

12. a. Given that $\sin\ \theta\ =\ \frac{1}{3}$, find $\cos\ \theta$, leaving your answer in surd form.

(5 marks)

b. **Table 2** shows some of the values for equation $y = x^3 - 2x^2 - 5x + 6$.

(i) Copy and complete the table.

Table 2 : $y = x^3 - 2x^2 - 5x + 6$

x	-3	-2	-1	0	1	2	3	4
y	-24		8		0	-4	0	18

(ii) Using a scale of 2 cm to represent 1 unit on the x-axis and 2 cm to represent 5 units on the y-axis, draw the graph of $y = x^3 - 2x^2 - 5x + 6$.

(iii) Use your graph to solve the equation $x^3 - 2x^2 - 5x + 4\ =\ 0$. **(10 marks)**

END OF QUESTION PAPER

NB: This paper contains 6 pages.